

Face Recognition Using Eigenfaces and Pattern Classification

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1 Problem and dataset

Closed-set face identification is the task of identifying the person in an image from a fixed set of enrolled individuals. Posed directly in pixel space the problem is badly conditioned: a 64×64 greyscale image is a point in $\mathbb{R}^{4096}$ and the dataset used here contains 400 such vectors, so features outnumber observations by about ten to one. The sample covariance matrix is singular by construction and no per-class density is estimable.

Face images do not fill $\mathbb{R}^{4096}$. They concentrate near a much lower-dimensional structure, because faces have a common geometry. The Eigenfaces method (Turk & Pentland, 1991) is built on this. The linear subspace holding most of the variation across a set of faces is estimated, and classification takes place inside it. That pipeline is implemented here from first principles, including the principal component analysis itself. The central question of the project is:

How much dimensionality can be removed from facial images before identity-discriminating information is significantly lost?

Two criteria give different answers. Under *reconstruction*, what counts is how much pixel variance is retained; under *discrimination*, how well identities can still be told apart. The main finding here is that the two criteria differ by a factor of about four, for a reason visible in the component structure.

The Olivetti (ORL) database (Samaria & Harter, 1994) consists of 400 greyscale images of 40 subjects, ten each, normalised to $[0, 1]$. Within a subject the images vary in lighting, expression, and the presence of glasses. All are cropped, frontal, and roughly aligned against a dark background, so the experiments isolate the recognition method from any face *detection* front end. The redundancy that motivates the method is measurable. Among 300 sampled pixels the mean absolute pairwise correlation is 0.26 and 10.4% of pairs exceed $|r| = 0.5$.

2 Eigenface method

2.1 Principal components

Each image is flattened row-major into $x_i \in \mathbb{R}^d$, $d = 4096$, and stacked into $X = [x_1, \dots, x_N]^T$ with $N = 400$. Writing $\mu = \frac{1}{N} \sum_i x_i$ and $A = X - \mathbf{1}\mu^T$, the sample covariance is

$$\Sigma = \frac{1}{N-1} \sum_{i=1}^N (x_i - \mu)(x_i - \mu)^T = \frac{1}{N-1} A^T A. \quad (1)$$

Figure 1 shows μ and three of the mean-centred images. PCA maximises the projected variance $v^T \Sigma v$ subject to $v^T v = 1$. With a Lagrange multiplier, $\mathcal{L} = v^T \Sigma v - \lambda(v^T v - 1)$, and $\partial \mathcal{L} / \partial v = 0$ yields the eigenvalue problem

$$\Sigma v_i = \lambda_i v_i. \quad (2)$$

By Equation (2), the stationary points of the projected variance are the eigenvectors of the covariance Σ in Equation (1), and since $v_i^T \Sigma v_i = \lambda_i$, each eigenvalue is the variance along the corresponding eigenvector. Keeping the top k eigenvectors as columns of W_k gives the rank- k subspace of least expected squared reconstruction error (Eckart & Young, 1936). Σ is real symmetric, so $W_k^T W_k = I_k$.

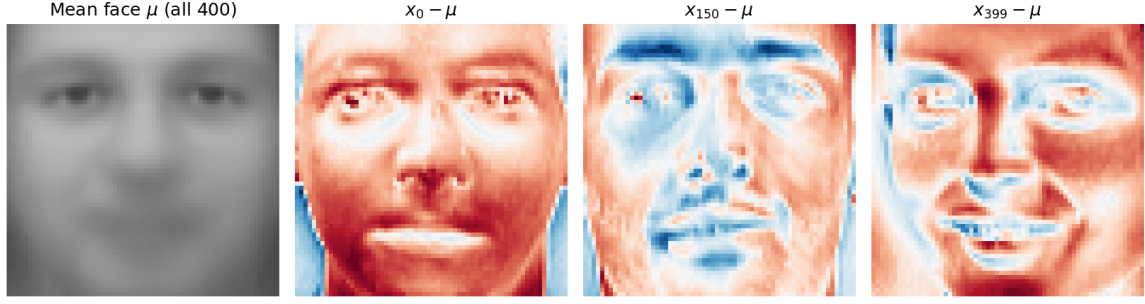


Figure 1: The mean face μ over all 400 images, and three mean-centred images $x_i - \mu$ (red above the mean, blue below).

2.2 Snapshot trick

Σ is 4096×4096 and singular: it sums N_{train} rank-one terms under one linear constraint, so $\text{rank}(\Sigma) \leq N_{\text{train}} - 1 = 279$ and at most 279 of its 4096 eigenvalues are non-zero. Take instead the Gram matrix $G = \frac{1}{N-1} AA^T \in \mathbb{R}^{N \times N}$. If $Gu_i = \lambda_i u_i$, left-multiplying by A^T gives

$$\frac{1}{N-1} A^T A A^T u_i = \lambda_i A^T u_i \iff \Sigma(A^T u_i) = \lambda_i (A^T u_i), \quad (3)$$

so $A^T u_i$ is an eigenvector of Σ with the same eigenvalue. By Equation (3), the non-zero spectrum of a 4096×4096 matrix follows from a 280×280 eigenproblem. On the machine used here, fitting took 0.021 s by this route against 12.7 s for the direct decomposition, a 614-fold reduction. The resulting vectors are normalised to unit length.

Each v_i lies in $\mathbb{R}^{4096}$, the space of the images themselves, so it reshapes to 64×64 and displays as a *pattern of deviation from the mean face*. As a linear combination of training faces, $v_i \propto \sum_j u_{ij}(x_j - \mu)$, it is itself face-like, hence the name, and a face is then represented by k coefficients over that shared basis: $x \approx \mu + \sum_{j \leq k} z_j v_j$ with $z = W_k^T(x - \mu)$.

2.3 Classification and verification

A test face is projected to $z = W_k^T(x - \mu)$ and classified by k -nearest neighbours (Duda et al., 2001). The non-parametric rule follows from the sample size. With 40 classes and 7 training images each, a full Gaussian per class would have $k + k(k+1)/2$ parameters to estimate from 7 observations. Because W_k is orthonormal, distances in the projected space equal distances between the pixel-space reconstructions. Two variants recur: *plain* projection, where components keep their natural scale so high-variance directions dominate, and *whitened* projection $\tilde{z}_j = z_j / \sqrt{\lambda_j}$, which rescales every component to unit variance and amounts to a Mahalanobis distance inside the retained subspace.

Checked against the exact SVD solver in scikit-learn, the implementation gives a maximum relative eigenvalue error of 5.0×10^{-15} , a minimum absolute eigenvector dot product of 1.0000000000, and a maximum reconstruction difference of 1.6×10^{-14} .

3 Experimental design

Each subject contributes n_{train} images to training and the rest to test, since a uniformly random split would leave some subjects with no training images and so unidentifiable by construction. The default is 7 train / 3 test per subject: $N_{\text{train}} = 280$, $N_{\text{test}} = 120$.

μ and W come from training images only, and test images are projected using those fixed parameters, because fitting PCA on the full 400 images before splitting leaks information into the basis and inflates the accuracy reported from it. With 120 test images a single split has a standard error near 0.024, so every figure reported here is the mean over 10 independent splits and is given with its standard deviation.

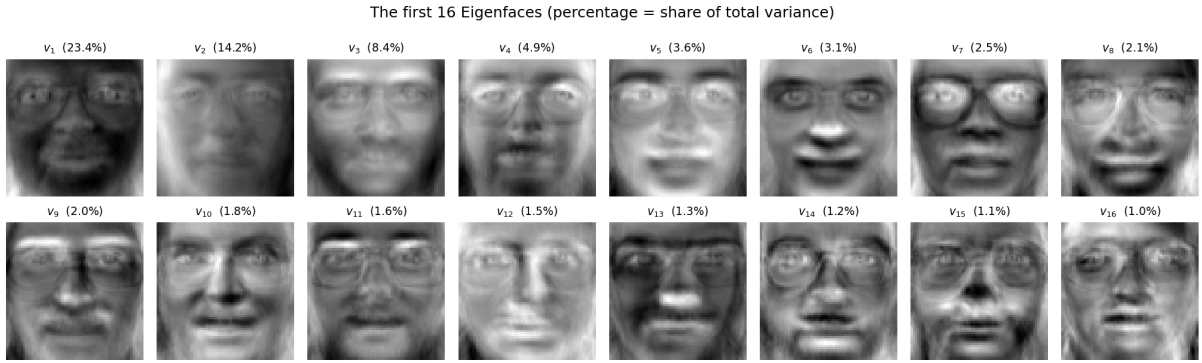


Figure 2: The first 16 Eigenfaces, with the share of total variance for each component.

4 Recognition results

The first three components in Figure 2 are low in spatial frequency and, on inspection, correspond to whole-image photometric structure rather than facial detail. v_1 is a broad contrast pattern separating the brow and eye band from the rest of the face, v_2 is nearly flat and encodes overall luminance, and v_3 is a top-to-bottom intensity gradient. None of the three distinguishes one person from another. Recognisable structure begins at v_4 : spectacle frames, hairline, moustaches, and changes around the mouth. Later components become more localised and higher in frequency, with identity information alongside sampling noise specific to these training images.

Meeting the variance thresholds set in the brief takes 59, 103, and 198 components for 90%, 95%, and 99% respectively, that is 1.44%, 2.51%, and 4.83% of the original 4096 dimensions. The share of total variance is 23.4% for the first component alone and 66.1% for the first ten.

In Figure 3 and Table 1 the two curves differ in shape, and the contrast between them is the clearest result here.

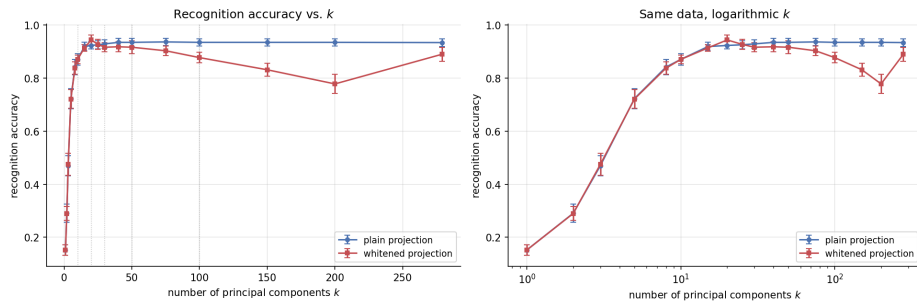


Figure 3: Recognition accuracy against k , plain and whitened projections, mean $\pm$ s.d. over 10 splits.

Table 1: Recognition accuracy against the number of retained components.

k	Plain	Whitened	k	Plain	Whitened
10	0.871 ± 0.021	0.871 ± 0.017	50	0.935 ± 0.015	0.917 ± 0.025
20	0.923 ± 0.014	0.944 ± 0.019	75	0.937 ± 0.013	0.903 ± 0.018
25	0.926 ± 0.016	0.928 ± 0.020	100	0.935 ± 0.013	0.878 ± 0.020
30	0.929 ± 0.016	0.917 ± 0.017	200	0.935 ± 0.013	0.778 ± 0.036

The plain curve rises and then flattens without falling, which follows from the geometry of the projection. W is orthonormal, so with every component retained the Euclidean distance in Eigenface space equals the pixel distance exactly, and adding components moves 1-NN monotonically towards the full-dimensional answer. Trailing components have tiny eigenvalues and contribute almost nothing to a Euclidean distance, so under this metric surplus components cost only computation.

Under whitening the curve rises, peaks at $k = 20$, then falls from 0.944 to 0.778 at $k = 200$. Whitening divides component j by $\sqrt{\lambda_j}$, which for a trailing component means dividing by a very small number and so scaling it up to the same variance as the leading ones. The i -th eigenvalue, however, comes from 280 observations in 4096 dimensions. For large i that estimate is increasingly dominated by sampling noise, and the direction then

reflects quirks of the particular training images more than a property of the population. Whitening amplifies exactly that noise, producing the familiar overfitting signature.

The useful number of components is set by how many directions are estimated reliably from the sample, not by how many exist mathematically, and whether the unreliable ones degrade accuracy depends on whether the metric gives them weight. Both curves follow from this. It is why they diverge, and why “how many components should I keep?” has no answer from the representation alone.

At $k = 75$, 1-NN with Euclidean distance gives accuracy 0.9367 ± 0.0137 , macro precision 0.9507, macro recall 0.9367, and macro F_1 0.9339. Macro-averaging weights every subject equally, so failure on a few individuals is not masked by a good average. On the reference split 8 of 120 test images are misclassified and the confusion matrix is close to diagonal, with no pair of subjects mutually confused.

Across distances at $k = 75$ and 1-NN, Manhattan distance gives 0.9442, Euclidean 0.9367, and cosine 0.9300, a margin well under one standard deviation. The effect of neighbourhood size is larger than that of the distance function. Euclidean accuracy falls to 0.854 at 3-NN and 0.805 at 5-NN. With 7 training images per subject a 5-NN vote often includes images of other identities from neighbouring clusters, so the usual argument that larger n smooths noise rests on a sample density absent here.

4.1 Reconstruction fidelity

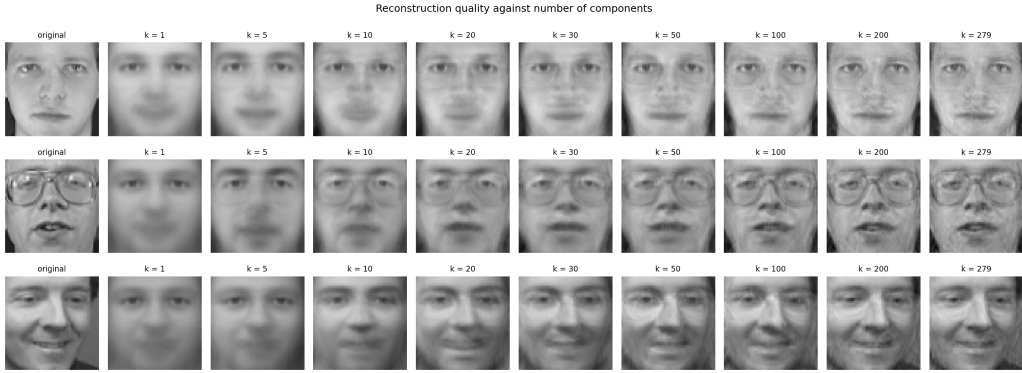


Figure 4: Reconstructions of three test faces at increasing k .

Reconstruction maps a projection back by $\hat{x} = \mu + W_k W_k^T (x - \mu)$, and since $W_k W_k^T$ is an orthogonal projector, the residual lies wholly in the discarded directions, giving $\mathbb{E}\|x - \hat{x}\|^2 = \sum_{j>k} \lambda_j$. In Figure 4, at $k = 1$ each face is the mean face with a photometric adjustment. Identity becomes visually recognisable around $k = 20$ to 30, and past $k \approx 100$ the changes are confined to fine texture. Test RMSE falls from 0.083 at $k = 10$ to 0.050 at $k = 100$ and 0.041 at $k = 279$, with PSNR 21.8, 26.2, and 28.1 dB.

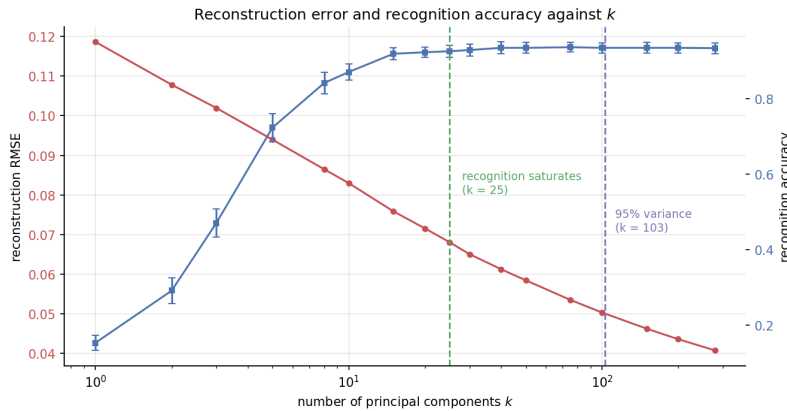


Figure 5: Reconstruction error and recognition accuracy against k .

Figure 5 shows reconstruction error still falling well past the point where recognition stops improving, because the variance removed across that range, fine skin texture, sensor noise, small expression detail, contributes to pixel fidelity and not to identity. Holding 95% of variance costs about four times the components required for recognition, $k = 103$ against $k = 25$, and moves accuracy from 0.926 to 0.935 (measured at $k = 100$), a

difference inside the split-to-split noise. A variance threshold is therefore the wrong criterion for choosing k in this recogniser: it is a measure of how well the subspace reproduces images, whereas recognition depends on how well the subspace separates identities.

5 Critical analysis

5.1 Enrolment size

Enrolment set size is the strongest single influence measured. Accuracy runs 0.547, 0.698, 0.774, 0.820, 0.867, 0.923, and 0.950 for 1, 2, 3, 4, 5, 7, and 9 training images per subject, and has not saturated at nine. Two mechanisms compound. Few reference images are enrolled per subject, so a test image with an unseen expression has no near neighbour of the correct identity; and the basis itself degrades, since at $n_{tr} = 1$ only 39 components exist, estimated from 40 images. The fall from 0.950 at nine images to 0.547 at one is the familiar one-sample-per-person difficulty.

5.2 Illumination mismatch

Table 2 and Figure 6 report accuracy when test images are multiplied by a left-to-right linear intensity ramp of increasing strength, modelling a directional source absent at enrolment, with training images untouched.

Table 2: Recognition accuracy under increasing illumination mismatch.

Ramp strength	0.0	0.2	0.3	0.4	0.5	0.7
Euclidean	0.937	0.878	0.758	0.488	0.305	0.142
Cosine	0.930	0.869	0.726	0.471	0.308	0.181
Drop first 3	0.932	0.938	0.936	0.920	0.890	0.783

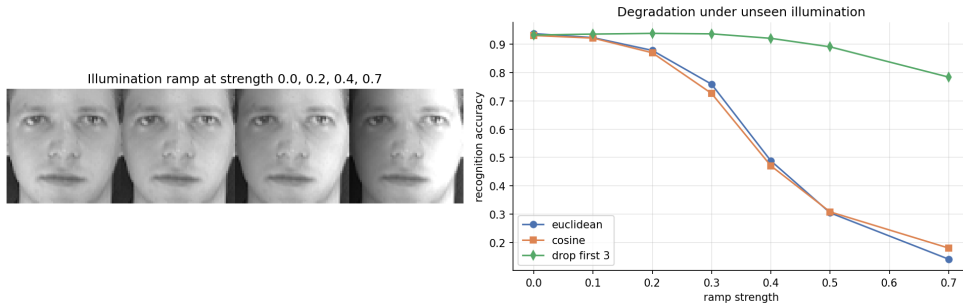


Figure 6: Left: the ramp at strengths 0.0, 0.2, 0.4, 0.7. Right: accuracy under increasing illumination mismatch.

Euclidean 1-NN falls from 0.937 to 0.142 across the ramp. Of the two mitigations tested, only one works.

Cosine distance gives no benefit: at each of the six ramp strengths its accuracy is within 0.04 of the Euclidean value. The expectation that it would help rested on a uniform brightness change, which shifts a projection mainly in magnitude. This ramp varies spatially, brightening one side of the face and darkening the other, so it rotates the projection without lengthening it, and discarding magnitude discards nothing relevant. Invariance to global scaling is of little use against lighting with spatial structure, which real lighting generally has.

With the first three components removed, accuracy under the strongest ramp is 0.783, against 0.142 for Euclidean 1-NN with those components, so those three directions contain nearly the whole perturbation. This is experimental support for the reading of the component structure given earlier: the leading eigenfaces encode photometric conditions rather than identity. Discarding the three directions with the largest share of variance in the entire dataset improves recognition under illumination change. Variance ranking and discriminative value differ, and at the top of the spectrum they oppose each other.

Under matched illumination, dropping up to three components moves accuracy by less than one standard deviation either way, and dropping beyond about five costs real accuracy. Removing three discards 46% of total variance and leaves recognition unchanged. Discarding leading components is therefore insurance against mismatched conditions, not a free gain.

6 Research question

1-NN on the raw 4096 pixels, with no reduction, gives 0.9342 ± 0.0142 , the reference against which Table 3 sets three tolerance levels.

Table 3: Dimensionality that can be removed at three tolerance levels.

Criterion	k	Accuracy	Dimensions removed	Variance retained
Matches peak	75	0.937	98.17%	92.4%
Within 1% of peak	30	0.929	99.27%	82.7%
Within 5% of peak	15	0.919	99.63%	72.7%

About 99% of the original dimensions can be discarded with no measurable loss of identity information. At $k = 25$ the Eigenface representation scores 0.9258 ± 0.0160 against the raw-pixel 0.9342 ± 0.0142 , nominally lower by well under the split-to-split noise, so the two are indistinguishable at this sample size. The 164-fold compression costs nothing in accuracy and gains nothing either.

Measured end to end, the Eigenface pipeline runs slower than raw 1-NN, 16.4 ms against 3.3 ms per split, because it includes fitting PCA on every split. The benefit of the reduction is at query time and at scale: each comparison costs 25 multiplications instead of 4096 and the gallery is smaller by the same factor, but across 400 images that saving does not cover the one-off decomposition.

Three qualifications apply. The bound comes from sample size, not from faces; the training set yields at most $N_{\text{train}} - 1 = 279$ components, so achievable compression depends on how many images were available rather than on any intrinsic dimensionality of human faces. The threshold depends on the criterion, since under reconstruction the answer is several times larger. It depends on operating conditions too, because every figure here comes from images captured much like the training images. A compression lossless under matched conditions need not stay lossless under mismatched ones.

7 Scope and limitations

ORL faces are cropped, frontal, aligned, and captured against a constant background in one laboratory, with no clutter, occlusion, or age variation. No detection or alignment stage was required. In deployed systems alignment error is often the dominant failure source, so these accuracies do not estimate unconstrained performance.

PCA is unsupervised and maximises total variance without reference to labels, which is consistent with the leading components encoding illumination. Fisherfaces (Belhumeur et al., 1997), maximising between-class to within-class scatter, typically does better under lighting variation. That comparison was not implemented here, however, so the claim that PCA is suboptimal rests on the observed component structure and the illumination experiment rather than on a direct measurement.

The model is linear, while pose and expression move a face along a curved manifold a linear subspace only approximates. The scale is small. Even across ten splits the standard deviation at the operating point is about 0.014, so differences of 0.01 or 0.02 have no interpretation here. Accuracy also falls as the gallery grows, so these figures would not transfer to thousands of identities. The illumination model is synthetic, omitting cast shadows, specularities, and saturation, all non-linear effects likely to degrade performance further. The setting is closed-set: nearest-neighbour classification always returns some identity, whereas a deployed system requires a rejection threshold, a separate problem.

8 Summary and conclusions

Recovering the eigenfaces from a 280×280 eigenproblem instead of a 4096×4096 one ran 614 times faster on the machine used and matched a reference implementation to machine precision. The leading eigenfaces encode photometric conditions rather than identity, a consequence of ranking directions by variance and supported by the illumination experiment. Recognition accuracy saturates at a few tens of components while reconstruction error keeps falling, so the conventional 95% variance threshold corresponds to roughly four times the components required for recognition. Retaining more components does not necessarily generalise better, and whether it degrades accuracy depends on the metric: plain Euclidean 1-NN plateaus, where a whitened metric degrades as noise-dominated components are amplified.

Enrolment set size has the largest effect of the factors tested, moving accuracy from 0.950 with nine training images per subject to 0.547 with one. Unseen illumination is the most damaging perturbation, and the effective remedy is to discard leading components, not to change the distance measure. Cosine distance, often recommended for this purpose, gave no benefit against spatially varying lighting.

All results come from `notebooks/eigenfaces_project.ipynb`, which runs top to bottom, writes every figure to `figures/` and every number to `report/results.json`, and uses fixed seeds so the reported accuracies reproduce exactly; the timings are wall-clock measurements from one machine.

References

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